

Assignment 1

Jason Huang

1. Question 1

- a. With mean = 0.001104, variance = 0.000096, skewness = -0.668831, excess kurtosis = 2.339849, Normal Distribution is ruled out as it needs both skew and kurtosis to be always 0. Lognormal is ruled out as well since it will result in always a positive skew, where here we have a negative skew. Student t is also ruled out because its skew is always symmetrical, 0, whenever its undefined where mine isn't. That leaves us with NIG, where the skew lets us take either positive or negative sign while kurtosis is always positive matching what we have there.
- b. 1% quantile from fitted normal = -0.021707, if the data should be normal, at 1% of 1000, there should be about 10 to fall below the 1% quantile. actual observations below the 1% quantile: 26, which means the data has a heavier left tail than normal distribution which lines up with the negative skewness and positive kurtosis found in part a.
- c. A fitted normal is symmetric with kurtosis of 0, the data has negative skew and kurtosis so the errors show up specifically in the tails. Negative skew shows there is a longer left tail where larger kurtosis means both tails are fatter than normal predict. In part B we saw that in the 1% quantile there is 26 instead of 10, so the error direction is that normal model understates downside/tail risk.

2. Question 2

- a. The scatter shows a clear linear relationship with a few points deviate much further from the trend line, which suggest the error term has fatter tails than normal distribution, student's t error since a normal error would rarely produce such large deviations.
- b. The violated assumption is normality of the error term. OLS doesn't need the normal error to be unbiased or consistent, it only needs the errors to average to zero and be uncorrelated with x, so I predict B to stay roughly the same across all three models. The real damage I would say is to the standard error, since its built on squared residuals, a few huge residuals will affect the average to make it out of proportion. I expect the reported SE to overstate the slopes true uncertainty and make any confidence interval or t-test built on it unreliable
- c. OLS beta = 2.9908, Normal MLE beta = 2.9908 (identical, since OLS and Normal MLE are the same estimator), t MLE beta = 3.0192. The difference is

about 0.03, less than 1%. This matches what I predicted that the slope barely moved.

- d. The AICc gap is about 27 points (692.145 vs 665.327), a wide margin favoring t, but the betas above are nearly the same. So the Normal model isn't wrong about the relationship between x and y, it's wrong about the error distribution around that relationship. It's underestimating how often large residuals occur, understating tail risk in the errors, not the trend itself.
- e. Normal's 95% quantile (2.2117) is wider than t's (2.0538), but at 99.5% they flip, t (4.6201) is much wider than Normal (3.4635). This is because the t distribution has thinner shoulders but fatter tails, near the middle it's actually tighter than normal, but far out its slower tail decay lets it keep assigning real probability where normal has already dropped to near zero. For a capital buffer, I'd want the t's 99.5% quantile, since a buffer is meant to cover extreme/rare losses, and using the Normal's number here would understate the capital needed for a true tail event.

3. Question 3

- a. X1 and x2 is a clear S shaped curve, not random. The curve is always increasing but not straight line, so Pearson should understate this relationship because it only measures straight line fit, while spearman should read it as nearly perfect since it only cares about rank order. Any pair with x4 looks like there's no pattern so I expect Pearson and Spearman to agree there.

99.5%: Normal 3.4635, t=4.6201

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=== Pearson ===
      x1      x2      x3      x4
x1  1.0000  0.7991  0.7195 -0.0047
x2  0.7991  1.0000  0.5876 -0.0209
x3  0.7195  0.5876  1.0000 -0.0086
x4 -0.0047 -0.0209 -0.0086  1.0000

=== Spearman ===
      x1      x2      x3      x4
x1  1.0000  0.9719  0.6839  0.0002
x2  0.9719  1.0000  0.6674 -0.0011
x3  0.6839  0.6674  1.0000 -0.0024
x4  0.0002 -0.0011 -0.0024  1.0000

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- b. Largest gap is x1 vs x2, Pearson = 0.7791 & Spearman = 0.9719 Gap = 0.1728
- c. Pearson measures how well a straight line fits the data, with the S curve, points fall off the straight line even though they never reverse directions, so Pearson gets penalized for the curve. Spearman on the other hand converts

everything to ranks first, so it doesn't care about the curve shape at all, it only cares whether its consistently increasing, which it is. So Spearman is the honest description of how strong this relationship is, Pearson isn't wrong, its just answering "is this linear".

4. Question 4

- Block 1 = x_1 , block 2 = x_2 , $\text{Var}(x_2 | x_1) = E22 - E12^2/E11 = 4.1006 - 1.6952^2/1.09888 = 1.4853$. The factor is $(1-p^2) = 1 - 0.7986^2 = 0.3622$. Knowing x_1 shrinks your uncertainty about x_2 down to ~36% of what it was
- No it doesn't depend on the value of x_1 you observed. Looking at the first formula Var , it has no x_1 in it, its built from covariance matrix, comparing that to the conditional means formula below, which has $x_1 - u_1$.
- $E[x_2 | x_1] = u^2 + (E12/E11)(x_1 - u_1)$. Coefficient $E12/E11 = 1.5428$ is the OLS slope when you regress x_2 on x_1
- 93.5% of points fell inside the band, which is under the nominal 95%
- Coverage: 95.65% within 1sd, 85.79% between 1-2sd, 90.20% beyond 2sd.
- The scatter spreads out as x_1 moves away from 0 from either direction but the band stays a constant width the whole way across. The assumption that broke is the multivariate normal's conditional variance is supposed to be constant regardless of x_1 but the data actual spread grows when moving away from the center. The true conditional variance depends on x_1 even though the formula says it shouldn't. From part c, the linear form of the conditional mean is still true, what isn't is using one constant width band everywhere, it should widen as x_1 grows instead of staying flat

5. Question 5

- Partial ACF cuts off, it starts with a sharp spike at lag 1 & 2 then nothing significant after, ACF decays gradually instead of cutting off. Cutting off in Partial ACF plus decaying in ACF is the AR signature, MA would show the opposite, ACF cuts off, Partial ACF will decay. Since Partial ACF cuts off after lag 2, that points to AR(2)
- The rule is a lag is considered "significant" if its bar stick out past the shaded 95% confidence band which for $n = 500$, its about ± 0.088 , "cut off" means bars are significant up through some lag and then every later lag falls inside the band, here PACF is significant at lags 1 and 2 then inside band from lag 3 onwards
- AR(1): AICc = 1418.587, AR(2): AICc = 1372.438, AR(3): AICc = 1374.343, MA(1): AICc = 1389.084, MA(2): AICc = 1381.232, MA(3): AICc = 1374.117,
- Yes, AICc selected AR(2) matching the prediction from the ACF/Partial ACF prediction

- e. AR(3) fit is tiny, basically noise, AICc penalizes each extra parameter you add so its weighting if the additional coefficient explain enough new variance to be worth the penalty. Here it doesn't, so AICc rejects AR(3) in favor of the simpler AR(2). R^2 wouldn't make the same call because it would never go down when you add a parameter, it can only improve or stay flat even if the parameter is just fitting noise. It has no built in penalty for extra complexity and it will keep AR(3) or even further lags.